

Report of Attendance

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Computer Vision for Archaeology Summer School

Institute of Archaeology of the Czech Academy of Sciences (ARÚB) | Brno | 14–18 September 2026

1. Overview

This report summarises my participation in the Computer Vision for Archaeology Summer School held in Brno from 14 to 18 September 2026. The programme introduced the theoretical foundations and practical applications of computer vision, artificial intelligence and image analysis in archaeology. In recent years, the field of archaeology has evolved considerably; computer vision has enabled researchers to process and to analyze large volumes of data and information in an automated and less labor-intensive manner. This advancement has saved considerable time and human resources, while also creating new opportunities for archaeological research. This summer school demonstrated how these new capabilities can help archaeological research and showcased some of the used cases.

In general, the workshops followed a consistent structure, with theoretical instruction in the morning and practical exercises in the afternoon. The summer school lasted for a full working week, with each day beginning at approximately 09:30 and ending between 17:00 or 18:00. The Summer School took place at the research infrastructure of the Archaeological Information System of the Czech Republic (AIS CR), hosted by the Institute of Archaeology of the Czech Academy of Sciences (ARÚB) in Brno. The educational programme was funded by ATRIUM, and co-organized by MAIA.

2. Daily Schedule

On the first day of the Summer School, an introductory segment took place on computer vision in archaeology, explaining the theoretical foundations of the field. Participants were then divided into groups to discuss the datasets used in their research, the types of data they possessed, and how computer-vision technologies could help them address their research questions. This was followed by the practical part of the workshop, which we ran our first computer vision-language model in Google Colab. The model (*CLIP*) connected images with textual descriptions, which was a great introduction to how computer vision can help us label and categorize our photo dataset. Furthermore, we were subsequently introduced to the Python programming language and learned how various available tools and AI agents can support programming tasks.

On the second day, we learned how photographs of archaeological objects should be prepared for use in artificial-intelligence and computer-vision applications. The concept of annotation; the process of identifying and labelling objects within images was explained, along with common data formats such as COCO and YOLO. CVAT was also introduced as a tool for image annotation. During the practical session, participants carried out annotation exercises using CVAT and basically to train the model. We were also introduced to ArchaeoTag, a game-based application for recognizing archaeological objects, which was a nice addition. The

day concluded with presentations of the participants' projects and a discussion of their research and potential applications of computer vision.

The third day focused on the application of computer vision in microscopy. It was emphasized that the preparation and pre-processing stage of microscopic images was as important as the language model itself. As a result, during the practical session, we ran the provided code in Google Colab to observe how such an image-preparation workflow functions in practice and how the images change and become homogeneous. Furthermore, it was showcased how microscopic images can help us identify patterns in sediment samples or in other organic materials and to assign the images in a corresponding category or context.

On Thursday, the programme focused on using satellite imagery and remote-sensing techniques to analyze the Earth's surface. The fundamental principles of satellite remote sensing, the process of image acquisition, and the analytical possibilities offered by satellite data were presented. During the practical session, we applied computer-vision techniques to enhance the visibility of archaeological features and support their automated detection through cropmarks. This exercise was completed in Google Colab using Sentinel-2 satellite imagery which we run the code provided. In the afternoon, a presentation and a practical demonstration on how computer vision can contribute to the classification and clustering of rock art was also presented.

On Friday, a coin dataset was presented to demonstrate how different methods can be combined to analyze and identify features on ancient coins. The final method was inspired by astronomy and used spatial-pattern recognition to identify different coin dies. Centroids were generated over the relief features of the coins to detect patterns associated with particular dies. Even when the outlines of the features were not entirely clear, the centroids helped approximate their spatial arrangement and supported the classification of the coins. The programme concluded with a presentation on Hugging Face, a platform that hosts a wide range of artificial-intelligence models, followed by a discussion of its potential applications.

3. Participant Experience

The summer school provided a well-structured and intensive introduction to computer vision for archaeology. By the end of the programme, I had gained a clearer understanding of what computer vision is and how it functions, as well as how these methods may be adapted to different archaeological research questions. Overall, the training was academically relevant, practically useful and very well organised.

It is also important to mention that the organizers and members of the Institute were extremely friendly, supportive, and willing to explain difficult topics or assist participants whenever necessary. On the first day, I experienced problems with my computer and network connection. The organisers responded immediately and provided a computer to me so that I could participate fully in the practical session, while the technical issue with my own device was to be resolved. I particularly appreciated this support because it prevented any interruption to my training.

Activities like organised outings, guided visits and informal social events also took place. These activities created valuable opportunities for networking and potential future collaborations but also helped participants and instructors become acquainted and contributed to a friendly and collaborative atmosphere throughout the week. It was an incredible experience, and I am truly thankful to the organizers and to the contributors.